

Algorithms

Lecture #06

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Asymptotic Notation

Asymptotic Notation

Formally

$\Theta(g(n)) = \{f(n) \mid \text{there exist}$
positive constants c_1, c_2
and n_0 such that
 $0 \leq c_1g(n) \leq f(n) \leq c_2g(n)$
for all $n \geq n_0\}$

Your response at this point might be,
“I am sorry that I asked!”

Asymptotic Notation

Asymptotic Notation

- This is written as " $f(n) \in \Theta(g(n))$ "
- That is, $f(n)$ and $g(n)$ are *asymptotically equivalent*.
- This means that they have essentially the same growth rates for large n .

Asymptotic Notation - Example

Asymptotic Notation

For example, functions like

- $4n^2$,
- $(8n^2 + 2n - 3)$,
- $(n^2/5 + \sqrt{n} - 10 \log n)$
- $n(n - 3)$

are all asymptotically equivalent. As n becomes large, the **dominant** (fastest growing) term is some constant times n^2 .

Asymptotic Notation - Example

Asymptotic Notation - Example

- Consider the function
$$f(n) = 8n^2 + 2n - 3$$
- Our informal rule of keeping the largest term and ignoring the constant suggests that
$$f(n) \in \Theta(n^2).$$
- Let's see why this bears out formally.

Asymptotic Notation - Example

Asymptotic Notation - Example

We need to show two things for
 $f(n) = 8n^2 + 2n - 3$

- **Lower bound:** $f(n) = 8n^2 + 2n - 3$ grows asymptotically at least as fast as n^2 ,
- **Upper bound:** $f(n)$ grows no faster asymptotically than n^2 ,

Asymptotic Notation - Example

Asymptotic Notation - Example

Lower bound: $f(n)$ grows asymptotically at least as fast as n^2 .

- For this, need to show that there exist positive constants c_1 and n_0 , such that $f(n) \geq c_1 n^2$ for all $n \geq n_0$.
- Consider the reasoning

$$\begin{aligned} f(n) &= 8n^2 + 2n - 3 \geq 8n^2 - 3 \\ &= 7n^2 + (n^2 - 3) \geq 7n^2 \end{aligned}$$

Asymptotic Notation - Example

Asymptotic Notation - Example

- We implicitly assumed that $2n \geq 0$ and $n^2 - 3 \geq 0$
- These are not true for all n but if $n \geq \sqrt{3}$, then both are true.
- So select $n_0 \geq \sqrt{3}$.
- We then have $f(n) \geq c_1 n^2$ for all $n \geq n_0$.

Asymptotic Notation - Example

Asymptotic Notation - Example

Upper bound: $f(n)$ grows asymptotically no faster than n^2 .

- For this, we need to show that there exist positive constants c_2 and n_0 , such that $f(n) \leq c_2 n^2$ for all $n \geq n_0$.

- Consider the reasoning

$$\begin{aligned} f(n) &= 8n^2 + 2n - 3 \leq 8n^2 + 2n \leq 8n^2 + 2n^2 \\ &= 10n^2 \end{aligned}$$

Asymptotic Notation - Example

Asymptotic Notation - Example

- For this, we need to show that there exist positive constants c_2 and n_0 , such that $f(n) \leq c_2 n^2$ for all $n \geq n_0$.
- Consider the reasoning
$$\begin{aligned} f(n) &= 8n^2 + 2n - 3 \leq 8n^2 + 2n \leq 8n^2 + 2n^2 \\ &= 10n^2 \end{aligned}$$
- Thus $c_2 = 10$.

Asymptotic Notation - Example

Asymptotic Notation - Example

- We implicitly made the assumption that $2n \leq 2n^2$.
- This is not true for all n but it is true for all $n \geq 1$
- So select $n_0 \geq 1$.
- We thus have

$$f(n) \leq c_2 n^2 \text{ for all } n \geq n_0 .$$

Asymptotic Notation - Example

Asymptotic Notation - Example

- From lower bound we have $n_0 \geq \sqrt{3}$
- From upper bound we have $n_0 \geq 1$
- Combining the two, we let n_0 be the larger of the two: $n_0 \geq \sqrt{3}$.

Asymptotic Notation - Example

Asymptotic Notation - Example

- In conclusion, if we let $c_1 = 7$, $c_2 = 10$ and $n_0 \geq \sqrt{3}$, we have

$$7n^2 \leq 8n^2 + 2n - 3 \leq 10n^2$$

for all $n \geq \sqrt{3}$

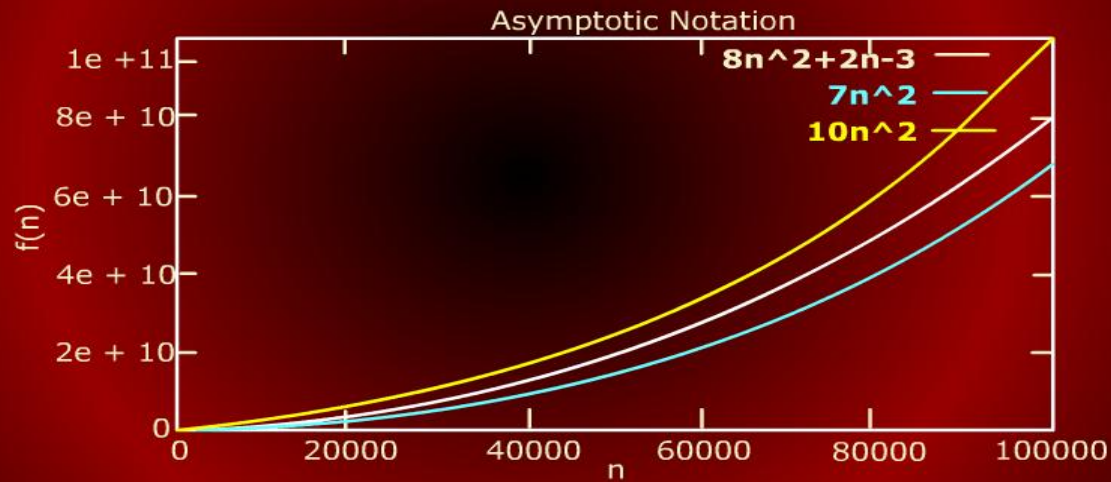
- We have thus established

$$0 \leq c_1g(n) \leq f(n) \leq c_2g(n)$$

for all $n \geq n_0$.

Asymptotic Notation - Example

Asymptotic Notation - Example



Asymptotic Notation - Example

Asymptotic Notation - Example

- We have established that $f(n) \in n^2$.
- Let's show why $f(n)$ is **not** in some other asymptotic class.
- First, let's show that $f(n) \notin \Theta(n)$.

Asymptotic Notation - Example

Asymptotic Notation - Example

show that $f(n) \notin \Theta(n)$.

- If this were true, we would have had to satisfy both the upper and lower bounds.
- The lower bound is satisfied because $f(n) = 8n^2 + 2n - 3$ does grow at least as fast asymptotically as n .

Asymptotic Notation - Example

Asymptotic Notation - Example

- Informally we know that $f(n) = 8n^2 + 2n - 3$ will eventually exceed c_2n no matter how large we make c_2 .
- To see this, suppose we assume that constants c_2 and n_0 did exist such that $8n^2 + 2n - 3 \leq c_2n$ for all $n \geq n_0$

Asymptotic Notation- Example

Asymptotic Notation - Example

- For this, need to show that there exist positive constants c_1 and n_0 , such that $f(n) \geq c_1 n^2$ for all $n \geq n_0$.
- Consider the reasoning
$$\begin{aligned}f(n) &= 8n^2 + 2n - 3 \geq 8n^2 - 3 \\ &= 7n^2 + (n^2 - 3) \geq 7n^2\end{aligned}$$
- Thus $c_1 = 7$.

Asymptotic Notation

Asymptotic Notation

- You may be asking that we continue to use the notation $\Theta()$ but have never defined it.
- Let's remedy this now.
- Given any function $g(n)$, we define $\Theta(g(n))$ to be a set of functions that *asymptotically equivalent* to $g(n)$.

2-Dimension Maxima Revisited

Design & Analysis of Algorithms

Asymptotic Notation Example

Asymptotic Notation - Example

- The lower bound is satisfied because $f(n) = 8n^2 + 2n - 3$ does grow at least as fast asymptotically as n .
- But the upper bound is false. Upper bounds requires that there exist positive constants c_2 and n_0 such that $f(n) \leq c_2n$ for all $n \geq n_0$.